

Customer Behavior Analysis Using SQL, Python and Power BI

Comprehensive Project Documentation

1. Project Overview

The objective of this project is to analyze customer shopping behavior and generate business insights that help organizations improve sales performance, customer retention, product recommendations, and marketing effectiveness. The project combines data cleaning, exploratory analysis, SQL-based querying, and dashboard development using Power BI.

2. Business Problem

Retail and e-commerce companies collect large volumes of customer transaction data. However, raw data alone cannot support decision-making. This project focuses on converting customer data into meaningful insights by identifying spending patterns, product preferences, customer demographics, seasonal trends, and satisfaction metrics.

3. Technology Stack

- Python (Pandas, NumPy, Matplotlib)
- SQL (MySQL or PostgreSQL)
- Power BI
- CSV Datasets
- Jupyter Notebook

4. Data Cleaning and Preparation

The first phase involves preparing the dataset for analysis. Typical activities include:

- Handling missing values and null records.
- Removing duplicate entries.
- Standardizing column names and data formats.
- Correcting inconsistent values.
- Converting categorical and numerical fields into usable formats.
- Validating data quality before analysis.

5. Exploratory Data Analysis (EDA)

EDA helps uncover trends and relationships within the data. Common analyses include:

- Distribution of customer ages.
- Purchase frequency analysis.
- Product category performance.
- Revenue contribution by customer segment.
- Seasonal purchasing trends.
- Customer rating analysis.

6. SQL Analysis

SQL is used to answer business questions efficiently. Example business questions:

- Which products generate the highest revenue?
- Which categories receive the best ratings?
- What is the average spending per customer?
- Which customer segments are most profitable?
- How does purchasing behavior vary across seasons?

7. Power BI Dashboard Development

An interactive dashboard is built to visualize KPIs and business insights. Recommended visuals include:

- Total Revenue KPI
- Total Orders KPI
- Average Customer Rating KPI
- Category-wise Sales Charts
- Age Group Analysis
- Seasonal Sales Trends
- Customer Loyalty Insights

8. Key Business Insights

Potential findings from the analysis may include:

- Certain product categories consistently outperform others.
- Younger and middle-aged customers contribute significantly to revenue.
- High-rated products often show better repeat purchases.
- Seasonal patterns influence demand and sales volume.
- Loyalty programs positively impact customer retention.

9. Business Recommendations

Based on analytical findings, organizations can:

- Improve inventory planning.
- Personalize product recommendations.
- Optimize marketing campaigns.
- Focus on high-value customer segments.
- Enhance customer satisfaction initiatives.

10. Skills Demonstrated

This project demonstrates the following professional skills:

- Data Cleaning and Transformation
- Exploratory Data Analysis
- SQL Query Development
- Dashboard Design and Visualization
- KPI Development
- Business Intelligence Reporting
- Problem Solving and Data Storytelling

11. Resume Description

Customer Behavior Analysis Project | SQL, Python, Power BI

- Performed end-to-end analysis of customer shopping behavior using Python and SQL.
- Conducted data cleaning, transformation, and exploratory analysis to identify business trends.
- Developed SQL queries to evaluate customer spending, product performance, and ratings.
- Built an interactive Power BI dashboard for KPI monitoring and business reporting.
- Generated actionable recommendations to support strategic decision-making.

12. Conclusion

This project demonstrates a complete analytics workflow from raw data processing to executive-level reporting. It showcases practical skills required for Data Analyst, Business Analyst, MIS Analyst, and BI Developer roles while providing measurable business value through data-driven insights.